

Employee Management App — Product Requirements Document

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Contributors: Engineering, HR Operations, Compliance **Created:** 2026-09-22 **Next Review:** 2026-10-06 **Classification:** Internal | Confidential

Intended Audience: | Reader | Sections of Primary Interest | |-----|-----|
Engineering | 6, 7, 8, 9, 10, 11, 12 | | Product / Design | 2, 3, 4, 5, 7, 8, 9 | | Executive / Stakeholder | 1, 2, 3, 5, 13 | | QA / Testing | 8, 9, 10 | | Security / Compliance | 10, 11, 12 |

1. EXECUTIVE SUMMARY

The Employee Management App is a multi-tenant B2B HRMS platform built to streamline workforce administration for modern businesses. Designed as a cross-platform solution with a React Native (Expo) mobile app for employees and a responsive Next.js/React web dashboard for HR and Admins, it eliminates manual timesheets, fragmented leave requests, and disparate payroll calculations. By integrating point-in-time geofenced attendance, multi-level leave approvals, and a highly configurable automated payroll engine into a single unified architecture (backed by Supabase), it empowers organizations to manage their teams with precision, compliance, and transparency.

This product needs to exist now to capture the growing mid-market demand for affordable, mobile-first HR tools that do not compromise on enterprise-grade security. Without a cohesive system, companies face severe compliance risks regarding location-tracking privacy, payroll inaccuracies, and scattered PII data. By architecting this V1 with strict tenant data isolation (Postgres RLS), time-expiring signed URLs for sensitive documents, and a localized payroll engine, we establish a secure foundation that can be commercialized to other businesses. Success for V1 means delivering a compliant, seamless HR experience that significantly reduces administrative overhead and completely eradicates attendance fraud (buddy punching).

2. PROBLEM STATEMENT

2.1 The Problem

Mid-sized businesses struggle with fragmented HR operations. Attendance is often tracked via honor-system spreadsheets or inflexible legacy biometric hardware that fails to accommodate

hybrid or multi-site workforces. Leave requests are managed through chaotic email threads, and payroll is calculated manually, leading to high administrative overhead, costly payroll errors, and an inability to accurately track field employee productivity. Furthermore, storing sensitive employee PII and payslips across unencrypted email servers or standard cloud drives presents massive regulatory liabilities.

2.2 Current Workarounds

Organizations currently use a mix of WhatsApp/Slack messages for leave approvals, Excel spreadsheets for manual timesheet tracking, and third-party accounting tools for isolated payroll. Field employees often call or text their managers to confirm they have arrived at a job site. This workaround costs HR departments hours of manual reconciliation every pay period, introduces a high margin of human error in tax and deduction calculations, and offers zero verifiable proof of an employee's physical presence on-site.

2.3 The Consequence of Inaction

If this product is not built, the business will remain bottlenecked by manual HR administration, effectively limiting the scale at which it can operate. Furthermore, lacking a verifiable, geofenced attendance system leaves the company vulnerable to "buddy punching" and time fraud, directly inflating payroll expenses. Commercially, failing to build a multi-tenant, compliant architecture now prevents the future SaaS monetization strategy.

2.4 The Opportunity

Solving this unlocks a scalable, automated HR engine. For the user, it means 100% transparency into their attendance, leaves, and salary via a consumer-grade mobile app. For the business, it unlocks real-time workforce visibility, automated compliance record-keeping, and hours of saved administrative labor each week. Architecturally, it creates a robust, secure B2B SaaS asset ready for market distribution.

3. GOALS & SUCCESS METRICS

3.1 Product Goals

ID	Goal	Type	Description
G-01	Eradicate Time Fraud	Business	Achieve 100% verifiable attendance via point-in-time geofencing.

ID	Goal	Type	Description
G-02	Automate HR Workflows	User	Reduce the time taken to process monthly payroll and reconcile leaves by 80%.
G-03	Mobile Self-Service	User	Enable employees to perform all core HR functions (clock-in, leave application, payslip download) via mobile.
G-04	Strict Multi-Tenancy	Security	Ensure absolute data isolation between tenant organizations using Supabase RLS.

3.2 Key Results & Success Metrics

Metric	Baseline	Target	Timeframe	Measurement Method
Daily Active Users (DAU)	0	90% of active workforce	30 days post-launch	Supabase Auth logs
Geofence Success Rate	N/A	> 95% punches successful	60 days	DB Attendance logs
Payroll Processing Time	Days	< 2 hours per tenant	90 days	App analytics / HR feedback
Unauthorized Data Access	N/A	0 instances	Perpetual	Audit logs / Pentest

3.3 Non-Goals

Non-Goal	Rationale	Planned For
Recruitment / ATS	Out of scope for V1; focus must remain on managing <i>active</i> employees.	V2 / Phase 2

Non-Goal	Rationale	Planned For
Performance Reviews	Complex workflows that dilute the core value of attendance/payroll.	V2 / Phase 2
Continuous GPS Tracking	High battery drain and severe privacy/compliance risks.	Never (Geofenced only)
IP-Whitelisted Web Logins	Deferred to prioritize mobile geofencing for launch.	Phase 2

4. TARGET USERS

4.1 User Types & Roles

User Type	Description	Primary Goal	Technical Level
Super Admin	Platform owner (You)	Provision tenants, manage SaaS billing	High
Tenant Admin	HR Director at client company	Configure policies, oversee all org data	Medium
HR Manager	HR Operations	Run payroll, manage employee records	Medium
Line Manager	Department head / Team lead	Approve team leaves and regularizations	Low-Med
Employee	Standard worker	Clock in/out, view payslips, request leave	Low

4.2 Roles, Permissions & Access Control

Permissions enforced at the database level via Supabase Row-Level Security (RLS).

Role	Resource	Create	Read	Update	Delete	Notes
Super Admin	Tenants	✓	✓ All	✓	✗	SaaS platform mgmt
Tenant Admin	All Org Data	✓	✓ Tenant	✓	✓ Soft	Full access within <code>tenant_id</code>
HR Manager	Payroll/Leaves	✓	✓ Tenant	✓	✗	Cannot alter company policies
Line Manager	Team Data	✗	✓ Reports	✓ (Approve)	✗	View/approve for direct reports only
Employee	Own Data	✓ (Punches)	✓ Own only	✗	✗	View own profile, payslips

5. MARKET & BUSINESS CONTEXT

5.1 Target Market

Dimension	Description
Geography	Global (Customizable payroll engine adapts to local tax laws)
Industry / Vertical	Industry-agnostic (focus on field services, retail, corporate)
Company size	SMB to Mid-Market (50 to 1,000 employees)
Buyer vs User	HR Director / CEO buys; Entire workforce uses

5.2 Monetization Model

Model	Description	Free / Paid Boundary	Launch / Later
Manual SaaS Billing	Tenant admins are billed manually offline based on active employee headcount.	N/A	Launch
Automated Stripe Sub	Automated \$X/user/month billing tier.	Free trial -> Paid	V2

6. SCOPE

6.1 In Scope — This Version

ID	Item	Rationale
IS-01	Tenant & User Provisioning	Email invite + Google SSO for seamless onboarding.
IS-02	Employee Profiles	Core directory, role mapping, document storage (signed URLs).
IS-03	Geofenced Attendance	Point-in-time GPS validation against office radius (Mobile).
IS-04	Anomaly Workflow	Automated shift closures and Missed Punch regularization.
IS-05	Leave Management	Multi-level approval workflows (Manager -> HR).
IS-06	Configurable Payroll	Dynamic engine (Base + Allowances - Deductions) -> PDF generation.

ID	Item	Rationale
IS-07	Push Notifications	Real-time alerts for approvals and company announcements.
IS-08	RLS & Secure Storage	Supabase RLS and private buckets for PII and payslips.

6.2 Out of Scope — This Version

Feature	Reason Deferred	Planned For
Expense Claims	Scope control; secondary to core payroll.	Phase 2
IP-Whitelisted Logins	Focus strictly on GPS geofencing for launch.	Phase 2
Automated SaaS Billing	Manual billing is sufficient for early adoption phase.	Phase 2

6.3 Assumptions

ID	Assumption	Confidence	Impact if Wrong	Owner
A-01	Employees possess smartphones capable of running Expo/React Native apps with GPS enabled.	High	Inability to use core attendance feature.	PM
A-02	Point-in-time geofencing satisfies local labor laws without triggering continuous-track	High	Legal redesign of the app.	Legal

ID	Assumption	Confidence	Impact if Wrong	Owner
	ing privacy violations.			

7. USER STORIES

Priority coding: ● P0 — Launch-blocking. ● P1 — High value. Should ship with v1.0. ● P2 — Nice to have.

7.1 Authentication & Tenant Onboarding

ID	User Story	Priority	Linked Feature
US-01	As a Super Admin, I want to create a new Tenant organization so that a client can use the platform.	● P0	F-01
US-02	As a Tenant Admin, I want to invite employees via email or let them login via Google SSO so they can access the app.	● P0	F-01

7.2 Attendance & Geofencing

ID	User Story	Priority	Linked Feature
US-03	As an Employee, I want to clock in/out using my mobile app so my attendance is recorded.	● P0	F-02
US-04	As a System, I must validate the Employee's GPS coordinates against	● P0	F-02

ID	User Story	Priority	Linked Feature
	the Tenant's configured office radius (geofence) to prevent buddy punching.		
US-05	As an Employee, I want to submit a Missed Punch regularisation request if I forgot to clock out, so I get paid accurately.	 P1	F-02

7.3 Leaves & Payroll

ID	User Story	Priority	Linked Feature
US-06	As a Line Manager, I want to receive push notifications for leave requests so I can approve them promptly.	 P0	F-03
US-07	As an HR Manager, I want to run a payroll calculation that automatically factors in leaves and missed days, so I don't have to use Excel.	 P0	F-04
US-08	As an Employee, I want to securely download my PDF payslip via a time-expiring link so my financial data is safe.	 P0	F-04

8. FEATURE REQUIREMENTS

Feature F-01 — Tenant Provisioning & Authentication

Priority: ● P0 **Summary:** Multi-tenant onboarding via Supabase Auth (Email + Google SSO).

8.1 Functional Requirements

Req ID	Requirement	Priority	Notes
F-01-R01	The system must support login via Email/Password and Google OAuth.	● P0	Handled via Supabase Auth.
F-01-R02	The system must link every user to a specific <code>tenant_id</code> .	● P0	Foundational for RLS.
F-01-R03	The system must enforce Supabase RLS so users can only read/write data where <code>data.tenant_id = user.tenant_id</code> .	● P0	Multi-tenant security.

8.2 Acceptance Criteria

- **Happy Path:** A new employee receives an invite link, logs in with Google, and is routed to their specific company's dashboard.
- **Security:** RLS policies mathematically prove that Tenant A cannot query Tenant B's data via direct API calls.

Feature F-02 — Point-in-Time Geofenced Attendance

Priority: ● P0 **Summary:** Mobile GPS validation against predefined office coordinates.

8.1 Functional Requirements

Req ID	Requirement	Priority	Notes
F-02-R01	The system must capture device GPS coordinates exactly at the moment of clock-in and clock-out.	 P0	No continuous tracking.
F-02-R02	The system must calculate the distance between the device GPS and the configured Office GPS using the Haversine formula (or similar geospatial query).	 P0	
F-02-R03	The system must block the punch if the distance exceeds the configured radius (e.g., 200 meters).	 P0	
F-02-R04	The system must run a daily CRON job at 23:59 (Tenant Timezone) to close active shifts and flag them as ANOMALY_MISSED_PUNCH .	 P1	Required for data integrity.

8.2 Edge Cases & Error Handling

Scenario	Trigger	Expected System Behaviour	User-Facing Message
Out of Geofence	User tries to clock in from home	Block punch, log attempt in audit.	"You must be within 200m of the office to clock in."

Scenario	Trigger	Expected System Behaviour	User-Facing Message
GPS Disabled	User denies location permissions	Block punch.	"Please enable Location Services to mark attendance."
Missed Punch	User forgets to clock out	CRON closes shift at 23:59, flags anomaly.	Notification next morning: "You missed a punch. Please regularise."

Feature F-03 — Multi-Level Leave Management

Priority:  P0 **Summary:** Leave application and approval workflows.

8.1 Functional Requirements

Req ID	Requirement	Priority	Notes
F-03-R01	The system must support a 2-step approval workflow (Line Manager $\rightarrow$ HR).	 P0	Configurable per tenant.
F-03-R02	The system must deduct approved leaves from the user's annual balance.	 P0	

Feature F-04 — Configurable Payroll & Secure Payslips

Priority:  P0 **Summary:** Dynamic salary calculation and compliant PDF storage.

8.1 Functional Requirements

Req ID	Requirement	Priority	Notes
F-04-R01	The system must calculate final pay	 P0	

Req ID	Requirement	Priority	Notes
	based on: $\text{Base} + \text{Allowances} - (\text{Leaves} * \text{Daily Rate}) - \text{Deductions}$.		
F-04-R02	The system must generate a PDF payslip and store it in a private Supabase Storage bucket.	 P0	
F-04-R03	The system must deliver the PDF to the user exclusively via a signed URL that expires in 15 minutes.	 P0	Hard compliance requirement.

9. USER FLOWS

Flow UF-01 — Geofenced Clock-In

Actors: Employee, Mobile App (React Native), Supabase Edge Function **Preconditions:** User is logged in. Location services are permitted on OS level.

1. User taps "Clock In" on Mobile Dashboard.
2. Mobile App requests current GPS coordinates from device OS.
 - └ OS DENIES: Show error "Location required." -> END
 - └ OS ALLOWS: App captures (Lat, Long, Timestamp).
3. App sends payload to Supabase API.
4. Database/Edge Function compares coordinates to Tenant Office location.
 - └ DISTANCE > RADIUS (e.g., >200m): Return 403 Forbidden.
 - └ App displays "You are outside the office zone."
 - └ DISTANCE <= RADIUS: Return 200 OK, Insert Attendance Record.
5. App displays "Successfully Clocked In at [Time]."

10. NON-FUNCTIONAL REQUIREMENTS

10.1 Performance Requirements

Requirement	Target	Measurement Method	Priority
API response time (Geofence check)	< 800ms	Supabase Dashboard	 P0
PDF Generation Time	< 5 seconds	APM monitoring	 P1

10.2 Security Requirements

Requirement	Standard	Enforcement
Multi-Tenancy Isolation	Supabase Postgres Row-Level Security (RLS)	Enforced on every table using <code>tenant_id</code> .
Document Storage	Private buckets + Signed URLs (15m expiry)	Supabase Storage configuration.
Authentication	JWT with refresh token rotation	Supabase Auth.

10.3 Compliance & Legal Requirements

Global Privacy (GDPR / CCPA / Labor Laws)

Requirement	Applicable	Implementation
GPS Tracking Limitations	 Yes	Point-in-time only. Background tracking strictly disabled.
Data Retention: Attendance	 Yes	Kept active for 3 years, then archived.
Data Retention: Payroll	 Yes	Kept for 7 years (Tax compliance standard).
Right to erasure	 Yes	Admin capability to hard delete PII.

11. INTEGRATIONS

Integration	Purpose	Direction	Required / Optional	Failure Impact
Supabase	Auth, Postgres, Storage, RLS	In/Out	Required	 Critical
Google OAuth	SSO Authentication	Inbound	Required	 High
Expo Push	Mobile Notifications	Outbound	Required	 Medium

12. DATA REQUIREMENTS

12.1 Core Data Entities

Entity	Key Attributes	Relationships	PII
Tenants	id, name, office_lat, office_lng, radius	Has many Users, Policies	No
Users	id, tenant_id, role, name, email, manager_id	Belongs to Tenant	Yes
Attendance	id, user_id, tenant_id, punch_in, punch_out, status, lat, lng	Belongs to User	Yes
Payslips	id, user_id, tenant_id, month, pdf_url (private)	Belongs to User	Yes

13. MILESTONES & TIMELINE

ID	Milestone	Description	Target
M1	Infrastructure & RLS	Supabase setup, Auth, strict RLS multi-tenancy.	Week 2
M2	Mobile Geofencing	React Native app with point-in-time GPS validation.	Week 5
M3	Leave & Workflows	Multi-level approvals and notifications.	Week 7
M4	Payroll Engine	Configurable math engine and PDF generation.	Week 9
M5	UAT & Pentest	Compliance checks, QA, and security validation.	Week 11

14. OPEN QUESTIONS

ID	Question	Impact if Unresolved	Owner	Status
Q-01	Will the PDF generation run via Supabase Edge Functions (Deno) or a separate Node microservice?	Dictates backend architecture.	Architect	Open
Q-02	How will Tenant Admins bulk-upload historical employee data	Impacts launch friction.	Product	Open

ID	Question	Impact if Unresolved	Owner	Status
	during onboarding? (CSV parser needed?)			